

Polynomial equations



1 Solve the following equations:

a $(5x - 4)(x + 3)(x - 2) = 0$

b $(x^2 - 16)(x^2 + 12x + 36) = 0$

c $(x + 3)^2(x - 6)(5x - 2) = 0$

d $3x^3 = 5x^2$

e $x^3 - 125 = 0$

f $x^3 + 5x^2 - 50x = 0$

g $x^4 - 16x^2 = 0$

h $64x^4 - 81x^2 = 0$

2 Consider the equation $2x^5 + 36x^4 - 80x^3 = 0$.

a Rewrite the equation in factored form.

b Solve the equation.

3 Consider the equation $4x^3 - 12x^2 - 5x + 15 = 0$.

a Rewrite the equation as the product of a linear factor and a quadratic factor.

b Solve the equation.

4 It is given that $x = 5$ is a root of the equation $x^3 + x^2 - 21x - 45 = 0$.

a Use long division or otherwise to rewrite the equation as the product of a linear factor and a quadratic factor.

b Solve the equation for the other root(s).

Equations reducible to quadratics

5 For each of the following equations:

i Using the variable p , determine a substitution that will reduce the equation to a quadratic in terms of p .

ii Make the substitution, and solve the new equation for p .

iii Solve the original equation for x .

a $x^4 - 34x^2 + 225 = 0$

b $x^4 - 26x^2 + 25 = 0$

c $25x^4 - 65x^2 + 36 = 0$

d $81x^4 - 117x^2 + 36 = 0$

e $(2x + 4)^2 + 10(2x + 4) + 16 = 0$

f $(3x + 3)^2 + 21(3x + 3) + 108 = 0$

g $(4x + 8)^2 + 16(4x + 8) + 48 = 0$

h $(4x - 4)^2 + 2(4x - 4) - 63 = 0$

6 Solve the following equations:



a $x^4 - 5x^2 + 4 = 0$

b $x^4 - 20x^2 + 64 = 0$

c $x^4 - 49x^2 + 600 = 0$

d $x^4 - 34x^2 + 225 = 0$

e $(4x + 9)^2 + 13(4x + 9) + 42 = 0$

f $(8x + 5)^2 + 17(8x + 5) + 70 = 0$

g $3(9x + 10)^2 + 19(9x + 10) + 20 = 0$

h $2(8x - 3)^2 + 13(8x - 3) + 15 = 0$

7 Solve for the real roots of the following equations:

a $x^4 = 8x^2 + 9$

b $x^4 = 3x^2 + 4$

c $x^4 = 12x^2 + 64$

d $x^4 = 24x^2 + 25$

e $(x^2 - 2)^2 - 25 = 0$

f $(x^2 - 7)^2 - 64 = 0$

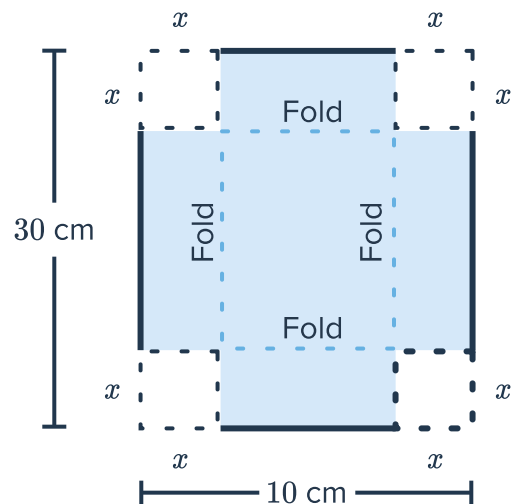
g $(x^2 - 5)^2 - 81 = 0$

h $(x^2 + 6)^2 - 15 = 34$

Applications

8 A box is formed by cutting squares of length x cm from the corners of a piece of cardboard 10 cm by 30 cm.

- a Find an expression for the volume of the box in terms of x .
- b Find the value of x if the volume is 176 cm^3 .



9 A retail outlet sells x units of a product each year. The annual cost for this product is given by $C = 9x + 3x^2$ and the annual gross profit is given by $G = 9 - 2x - 4x^2 + x^3$ where C and G are both in dollars.

- a Form an expression for P , the net profit after producing and selling x units.
- b Find the number of units that must be sold in order to make a profit of \$40 672.

- 10 A two-sided skate ramp is to have dimensions as shown in the image. One of the sloped sections is 50% longer (horizontally) than the other.



- a If the total volume of the two sloped sections is 240 ft^3 , construct an equation in terms of x to represent the situation.
- b Determine the dimensions of the skate ramp.

